Crossroads (Rural / Infrastructure / Multiple Routes & Strategic Control)

Executive Summary

Crossroads is a **rural-infrastructure battlefield** built around **multi-route maneuvering, strategic control of intersections, logistics warfare, and flexible deployment paths**. The terrain features rolling farmland, utility substations, rail spurs, fuel depots, bridges, overpasses, and a major highway interchange that forms the strategic heart of the map. The environment emphasizes Obsidian Protocol's autonomous route -selection logic, sensor-driven positioning, comms-relay networks, and logistics under open-terrain constraints. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of rural infrastructure. Key features include multi -route intersections, bridge chokepoints, rail-yard flanking paths, and strategic control of the central interchange.

Design Goals & Gameplay

Multi-Route Maneuver Warfare

Crossroads prioritizes flexible movement across multiple parallel and intersecting routes. Players must control intersections, bridges, and overpasses to dictate the flow of battle.

Strategic Infrastructure Control

Fuel depots, substations, and rail yards serve as critical strategic points. Holding them grants logistics and sensor advantages.

Autonomy & Route-Selection Logic

Autonomous AI chooses between highway lanes, rural roads, rail-side paths, bridge crossings, or off-road fields. Units adapt to blocked routes and enemy control of intersections.

Sensor & Information Warfare

Dust, smoke, and infrastructure clutter distort sensors. Relay towers on overpasses restore clarity. Destroying relays creates blind zones across the road network.

Logistics & Resources

Fuel depots, rail yards, and utility substations serve as resupply nodes. Convoys must navigate exposed stretches and avoid ambush -prone intersections.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between convoy escorts, fast recon units, or heavy armor. Blue deploys from the southwest farmstead; Red deploys from the northeast interchange command.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² rural zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Farmstead Command) — (0,0), 140×140 m Barns, fuel tanks, recon vehicles, rural staging area.
Red HQ (Interchange Command Node) — (1000,1000), 140×140 m Elevated highway command post with radar mast and AA emplacements.
Central Interchange Hub — (500,450), 300×250 m Highway overpasses, ramps, intersections, multi-lane combat zone.
Fuel & Logistics Depot — (250,800), 100×100 m Fuel tanks, trucks, pipelines, cargo crates.
Rail Spur & Yard — (800,300), 150×150 m Rail tracks, cargo cars, loading cranes, flanking routes.
Bridge Crossing — (0,450 → 400,900) Critical chokepoint: river + bridge + roadside cover.
Rural Road Route — (300,0 → 450,350) Safer but slower path with partial cover.
Overpass A — (650,150), +240 m Long-range sensor vantage point.
Overpass B — (150,650), +260 m Comms relay tower and recon hub.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Bridge Crossing

Major chokepoint; controlling it determines east-west movement.

Interchange Hub

Multi-lane combat zone; ideal for armor and ambushes.

Rail Spur

Provides stealthy flanking routes behind industrial cover.

Rural Road Route

Predictable but safe; vulnerable to overwatch from Overpass A.

Sightlines

Overpass A provides long-range sensor coverage (~350 m). Dust and smoke create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW farmstead + convoy staging.
Red Deployment: NE interchange command + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Valley floor, farmland, river, bridge, overpasses, rail yard.

Vegetation

Fields, shrubs, scattered trees, grass patches.

Buildings & Props

Military: Watchtowers, bunkers, radar dishes, AA guns. Industrial: Fuel tanks, pipelines, generators, cargo yards. Logistics: Trucks, crates, fuel drums, convoy vehicles.

Effects

Dust clouds, exhaust haze, smoke plumes, wind gusts.

Audio

Engine noise, wind, distant machinery, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Cliffs/Roc	3	200k	600k
Roads/Hi	4	80k	320k
Buildings	15	50k	750k
Bridges/C	4	30k	120k
Vegetatio	120	2k	240k
Props	50	10k	500k
Total	—	—	**~2.53M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out interchange, bridge, rail yard, AR anchor test.
2. **Core Systems — Month 2** NavMesh, route-selection logic, autonomy behaviors.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (dust, smoke).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify route-selection logic, convoy pathing, ambush detection.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, chokepoint behavior, convoy logic, deployment budget enforcement, anchor drift, sensor accuracy.